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Do Credit Rating Agencies Learn from the Options Market?

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Abstract. Do credit rating agencies (CRAs) learn from the options market? We examine this question by exploring the relation between options trading activity and credit rating accuracy. We find that as options trading volume increases, credit ratings become more responsive to expected credit risk and exhibit greater ability to predict future defaults. We also find that CRAs rely more on the options market as a source of ratings-related information when firm default risk is higher, options trading is more informative, manager-provided information is of lower quality, and firm uncertainty is higher. Our results are robust to a number of sensitivity tests, including alternative measures of options trading and credit rating accuracy. We reach similar inferences using various approaches to address endogeneity issues, including difference-in-difference analyses and an instrumental variables approach. Overall, our findings are consistent with the view that CRAs incorporate unique information from the options market into their rating decisions which, in turn, improves credit rating accuracy.

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Keywords: credit rating • credit risk • default probability • credit rating accuracy • options market

1. Introduction

High-quality credit ratings are crucial for the proper functioning of the financial system. In addition to bridging the information gap between firms and their capital providers, credit ratings facilitate debt contracting and regulatory compliance (Beaver et al. 2006, Kisgen 2006, Listokin and Taibleson 2010). Credit rating failures in the runup to the 2008–2009 global financial crisis highlight the importance of credit rating responsiveness to changing corporate and market conditions. In addition, the ability of credit ratings to predict future default probabilities is essential for an efficient allocation of capital across firms within the economy. Therefore, it is critical to understand the factors that influence the quality of credit ratings. Recent research shows that the options market contains unique information about implied volatilities and negative skews—particularly useful in assessing credit risk—which cannot be learned from the stock market (Xing et al. 2010, Jin et al. 2012). Although previous studies find that the information from the options market benefits investors and corporate managers (Johnson and So 2012, Blanco and Wehrheim 2017), no study to date has examined whether the options market provides informational benefits to credit rating agencies (CRAs). To fill the void in the literature, we explore whether CRAs learn new information from the options market to improve the accuracy of their credit ratings.

We focus on the options market for several reasons. First, options trading has grown considerably over the last two decades, with the total volume of equity options traded on U.S. exchanges increasing from 198 million contracts in 1999 to 405 million contracts in 2018.[1] A recent Goldman Sachs report shows that the average daily value of single stock option volumes surpassed that of stock trading volumes beginning in July 2020.[2] Second, unlike stock listings for which firms must actively apply, option listings are determined by exchanges and are therefore independent of firm decisions (Blanco and Wehrheim 2017). Third, the options market is well suited to informed investors exploiting their nonpublic information about future stock price volatility and negative "black swan" events (Jin et al. 2012), both of which are highly relevant to credit risk assessment.[3] The options market incorporates unique information about future stock price volatility and negative black swan events, and investors can extract such information by relying on implied volatilities and negative skews (i.e., smirks) from the options market that cannot be obtained from alternative sources, such as the stock or bond markets (Ni et al. 2008, Xing et al. 2010, Jin et al. 2012).

Given the significant informational role of the options market and the uniqueness of ratings-relevant information reflected in option prices, we posit that CRAs will access this rich source of information to improve their credit rating accuracy. CRAs have the incentive to enhance their reputations and profitability by providing more accurate credit ratings (House 1995, Meltz 2007, Pitman 2008). Because the options market reflects new information more rapidly than other markets (Xing et al. 2010, Patel et al. 2020) and provides forward-looking information that aligns well with the forward-looking nature of credit ratings, CRAs are expected to monitor and learn directly from the options market. CRAs might also learn indirectly through information spillovers from the options market into other markets (e.g., stock and bond markets). Therefore, we expect that CRAs will enhance their credit rating accuracy through direct (or indirect) learning from the options market. It is also possible, however, that CRAs do not use the options market as an important source of information due to incentive issues stemming from excessive market power (Partnoy 1999, 2006), commonly used issuer-pay models (Jiang et al. 2012), internal resource constraints, or privileged access to private information from the managers of rated companies (Bonsall 2014). Hence, whether CRAs learn useful information from the options market to improve rating accuracy is an open empirical question.

We address this question by examining the relation between options trading activity and credit rating accuracy using a large sample of publicly traded U.S. firms that have both credit ratings and listed options from 1996 to 2016. Following Roll et al. (2010) and Johnson and So (2012), we capture options trading activity for a given firm via the ratio of options trading volume to stock trading volume (*Vratio*).[4] We use two measures of credit rating accuracy that have been widely used in the literature (Becker and Milbourn 2011, Kedia et al. 2014, Bonsall et al. 2015, Badoer and Demiroglu 2018): the responsiveness of credit ratings to expected credit risk (i.e., rating-risk responsiveness) and the ability of credit ratings to predict future defaults (i.e., rating-default predictability).[5] These measures gauge how well credit ratings capture credit/default risk, which is the main purpose of credit ratings (Cantor and Packer 1995, Becker and Milbourn 2011).

Our results show that greater options trading activity is associated with more accurate credit ratings. Specifically, we find that as options trading volume (i.e., *Vratio*) increases, credit ratings become more responsive to expected credit risk and exhibit greater ability to predict future defaults.[6] These results hold after controlling for other factors that prior research has shown to influence credit risk (e.g., leverage, profit volatility). These results are also robust to alternative measures of options trading activity and credit rating accuracy and to alternative regression procedures.[7] Overall, the evidence is consistently supportive of our hypothesis that CRAs use information from the options market to improve the accuracy of their credit ratings.

One may argue that our results could be driven by some correlated omitted variables or reverse causality. To address these endogeneity concerns, we include firm fixed effects throughout our analyses to control for time-invariant, firm-specific unobservable or omitted factors; control for firm characteristics that prior studies identify as determinants of credit risk (e.g., leverage, profit volatility); and lag the options trading variable relative to the rating accuracy variable. We also conduct various cross-sectional analyses that help to mitigate omitted variables concerns.

Furthermore, we conduct three additional analyses. First, we perform a difference-in-differences analysis where we use a sample of firms with options listed for the first time and a matched control sample without listed options over the same period. We find that the mapping between the expected default probabilities and credit ratings becomes stronger after the option listing event for listed firms relative to nonlisted firms over the same period. Second, we use an instrumental variables approach using open interest and the lagged mean values of industry-quarter options trading volumes as instruments and continue to find similar results. Third, we control for potential firm-level characteristics that have been shown to be associated with options trading activity (Roll et al. 2010) and find similar results. All these tests help to corroborate our causal inference that greater options trading activity leads to more accurate credit ratings. Nevertheless, we also acknowledge that endogeneity issues and the potential for correlated omitted variables can never be completely ruled out.[8]

Finally, we investigate in greater detail the learning mechanism underlying the positive relation between options trading and rating accuracy. First, we examine the relation between options trading volume and credit rating sensitivity to implied volatility measures estimated exclusively from options prices. We find that when options trading volume increases, credit ratings become more sensitive to implied volatility levels, skews, and spreads. We also examine the relation between option trading volume and credit rating sensitivity to stock return volatility and bond yield spread. We find some evidence that as option trading volume increases, credit ratings become more sensitive to stock return volatility and bond yield spread. These results suggest that in addition to learning directly from the options market, CRAs might also learn indirectly through information spillovers from the options market into the stock and bond markets. Second, we use cross-sectional analyses to examine situations in which CRAs are more likely to seek information from the options market. We find that the impact of options trading on credit rating accuracy increases when firm default risk is higher, options trading is more informative, manager-provided information is of lower quality, and firm uncertainty is higher.[9] These results provide further support for the learning mechanism.

Our study contributes to the literature in two primary ways. First, we extend the literature examining the factors that influence credit rating quality. This stream of research has focused primarily on the effects of company-provided information while largely ignoring alternative information sources, such as the options market (Alissa et al. 2013, Bonsall et al. 2017, Zhang 2018). We extend this stream of research by showing that CRAs can use unique information from the options market to improve credit rating accuracy. To the best of our knowledge, we are the first to explore the incremental information impact of the options market on an intermediary which has privileged access to company-provided private information (Bonsall 2014). Although managers might be reluctant to convey negative private information to CRAs, the options market specializes in this very type of information. Our results can be useful to regulators interested in improving credit rating quality (Manns 2013, Stivers Slams Section of Financial Reform Law 2011, U.S. House of Representatives 2011).

Second, our study adds to the literature on the economic effects of the options market. Earlier studies explore the informational benefits of the options market to shareholders (Amin and Lee 1997, Easley et al. 1998, Pan and Poteshman 2006, Roll et al. 2009, Chan et al. 2016, Truong and Corrado 2014). Roll et al. (2009), for example, show that greater options trading activity corresponds to higher firm values, as reflected in Tobin's Q. Prior research, however, has devoted little attention to the impact of the options market on creditors in general and no attention to its impact on CRAs in particular. We fill this gap by showing that the unique information produced by option traders can be accessed by CRAs to generate more accurate credit ratings. This positive effect can be of interest to both options market participants and the Securities and Exchange Commission (SEC), which establishes options-listing eligibility criteria.[10]

The remainder of the paper is organized as follows. Section 2 presents the related literature and hypothesis development. Section 3 describes the sample, variable measurement, and empirical model. Section 4 presents the main results. Section 5 provides a discussion of underlying mechanisms and presents related tests, and Section 6 concludes the study.

2. Related Literature and Hypothesis Development

2.1. Informational Benefits of the Options Market

Numerous studies recognize that stock options encourage informed trading. The options market is potentially more attractive to informed traders than the stock market due to its lower transaction costs and higher leverage. In other words, the options market gives informed investors higher leverage at a lower cost for each dollar of investment (Black 1975, Cox and Rubinstein 1985, Back 1993, Biais and Hillion 1994). In addition, options traders do not face short-sale constraints and can use a wider range of investment strategies than can their counterparts in the stock market (Naiker et al. 2013). All these factors make it possible for informed investors to earn profits in the options market that are higher than those in the underlying (i.e., spot) stock market, which can encourage option traders to spend additional time, effort, and resources to generate unique, nonpublic information. As a result, the options market should help to promote higher informational efficiency.

Empirical evidence confirms the informational benefits of the options market. Easley et al. (1998) and Pan and Poteshman (2006) show that options trading volume imparts information on the direction of future stock prices. Chan et al. (2016) find that option trades incorporate information about future stock returns over a window of several months. Chakravarty et al. (2004) estimate that the options market contributes to approximately 17% of price discovery in the underlying stock market and find that options market price discovery is positively associated with options trading volume. Consistent with this evidence, Blanco and Wehrheim (2017) find that firms with higher options trading volume are more innovative because their higher information efficiency reduces information asymmetries inherent to research and development (R&D) decisions.

In addition to the stock price efficiency benefits, the options market provides unique information about a firm's implied volatility level, spread, and skew. Using option prices and contract-related inputs (e.g., exercise price, time-to-maturity), CRAs can extract estimates of a firm's future volatility and probability of default from option pricing models such as Moody's Analytics Kealhofer, McQuown and Vasicek (KMV) model. Such information is unavailable from the equity or bond markets. As Ni et al. (2008, pp. 1059–1060) explain, "Unlike traders with directional information about underlying stock prices who can trade in either the stock or options market, traders with volatility information can only use nonlinear securities such as options." Other studies show that negative news is more rapidly and accurately incorporated into option prices than into stock prices. Xing et al. (2010) show that (p. 641) "Informed traders with negative news prefer to trade out-of-the-money put options, and … the equity market is slow in incorporating the information embedded in volatility smirks." Consistent with the credit-risk relevant information in the options market, Cremers et al. (2008) show that option implied volatilities and skews have explanatory power for corporate bond spreads.[11]

2.2. Hypothesis Development

Our underlying hypothesis is that CRAs can use the unique information from the options market to improve their credit rating accuracy. CRAs have the incentive to enhance their reputations and profitability by providing more accurate credit ratings (House 1995, Meltz 2007, Pitman 2008).[12] Accordingly, we expect that CRAs will extract ratings-relevant information directly from the options market. One reason is that the information from the options market is forward-looking, which aligns well with the forward-looking nature of credit ratings (i.e., firm credit ratings reflect CRAs' assessments of future credit risk). Another reason is that the options market reflects new information more rapidly than other markets. Prior studies show that the options market leads the stock market in incorporating new information (Easley et al. 1998, Pan and Poteshman 2006, Ni et al. 2008, Patel et al. 2020), and the stock market leads the bond and Credit Default Swap markets (Downing et al. 2009, Zhang et al. 2009, Hilscher et al. 2015). Given this lead pricing role of the options market, CRAs might also learn indirectly through information spillovers from the options market into other capital markets.[13]

The idea of CRA learning is that CRAs may not have full information about all relevant factors of credit risk assessment, despite their access to private information from their corporate clients.[14] Corporate managers may be unable to provide certain ratings-relevant information to CRAs due to their lack of such information, corporate bureaucracy, or because doing so would be incompatible with managerial incentives (Rajan and Zingales 2003). Hence, the options market may collectively reveal ratings-related, nonpublic information that is not possessed by CRAs or provided by managers because it aggregates disparate pieces of information from many traders.[15]

Specifically, the information from the options market could be related to macroeconomic conditions, consumer demand, and industry competition (Bond et al. 2012, Zuo 2016). The options market also provides nonpublic information about future volatility and negative black swan events (Jin et al. 2012) that cannot be extracted from alternative sources (Ni et al. 2008, Xing et al. 2010, Jin et al. 2012). In short, CRAs can learn ratings-relevant information at the firm, industry, and macro level from the options market to improve their rating accuracy.

Because the informational benefits of the options market depend on the volume of options traded (Roll et al. 2009, Blanco and Wehrheim 2017), we posit that credit ratings will be more accurate when options trading volume is higher. We state our hypothesis in the alternative form as follows.

Hypothesis. *Greater options trading activity is associated with greater credit rating accuracy.*

Although there are strong (ex ante) reasons to expect that higher options trading volume will lead to more accurate credit ratings, the options market might have little (if any) impact on credit rating accuracy for at least four reasons. First, federal regulations have given substantial market power to CRAs designated as Nationally Recognized Statistical Ratings Organizations (Partnoy 1999, 2006), which can reduce their incentives to collect additional information from the options market. Second, the issuer-pay model could subject CRAs to conflicts of interest (Jiang et al. 2012), which might make them less responsive to market information. Third, insufficient staff and other binding resource constraints faced by CRAs (SEC 2012) could limit their ability to collect and process relevant information from the options market.[16] Fourth, because CRAs have privileged access to private information from client management (Bonsall 2014), they might not consider the options market as an important source of information; instead, CRAs might ignore options-related information if they believe that the private channel is sufficient. Overall, whether CRAs use information from the options market to improve their rating accuracy is an open empirical question.

3. Methodology and Sample

3.1. Measurement of Options Trading

Following Roll et al. (2010), we measure options trading activity using the ratio of options trading volume (aggregated across all calls and puts) to stock trading volume over the same fiscal quarter. We multiply this ratio by 100 to derive our measure of options trading activity, *Vratio*, as one exchange-traded option contract represents 100 underlying shares of stock. Because we deflate option trading volume by stock trading volume, this measure has the advantage of controlling for intrinsic trading intensity due to firm characteristics. This measure has been shown to exhibit cross-sectional and time-series variation that is reflective of informed trading (Roll et al. 2010), is higher around earnings announcements, and predicts postannouncement returns (Roll et al. 2010) and future firm-specific earnings news (Johnson and So 2012), suggesting that it contains value-relevant, nonpublic information.[17]

3.2. Empirical Model

To examine whether CRAs learn from the options market, we use two approaches to explore the effect of options trading on the accuracy of credit ratings. The underlying idea of both approaches is that options stimulate informed trades and that the informational benefits of options depend on trading volume (Blanco and Wehrheim 2017). The first approach explores how lagged options trading activity affects the contemporaneous responsiveness of credit ratings to expected credit risk as reflected in the expected default probability derived from the Merton (1974)/KMV model (i.e., rating-risk responsiveness). A number of studies demonstrate that the Merton/KMV model offers an unbiased and informative measure of expected credit risk and outperforms credit ratings that are issued by S&P and Moody's in reflecting a firm's credit risk (Arora et al. 2005, Korablev and Dwyer 2007). Not surprisingly, rating-risk responsiveness has been widely used in prior studies as a measure of credit rating accuracy, including Kedia et al. (2014), Xia (2014), Bonsall et al. (2015), Cornaggia et al. (2016), and Bonsall et al. (2017).

Specifically, we estimate the following equation via ordinary least squares (OLS):

$$Rating_{t+1} = \alpha_0 + \beta_1 EDP_{t+1} + \beta_2 Vratio_t + \beta_3 EDP_{t+1} \times Vratio_t + \sum_{q=4}^{m} \beta_q (q^{th}\,Control\ Variable_t) + \sum Firm + \sum Year + \varepsilon, \quad (1)$$

where $Rating_{t+1}$ represents a firm's numerical credit ratings from S&P that is averaged over fiscal quarter $t+1$, with higher values representing lower (or worse) ratings.[18] EDP_{t+1} represents the expected default probability for a firm in quarter $t+1$ derived from the Merton (1974) model. A stronger relationship between $Rating_{t+1}$ and EDP_{t+1} indicates that credit ratings are more responsive to expected credit risk and, hence, more accurate, which corresponds to a positive coefficient on EDP_{t+1} (i.e., $\beta_1 > 0$). Accordingly, our hypothesis anticipates a positive coefficient on the interaction of EDP_{t+1} with $Vratio_t$ (i.e., $\beta_3 > 0$), given that it predicts a positive association between options trading activity and rating accuracy.

Our second approach assesses rating accuracy based on the extent to which ratings predict future defaults (i.e., rating-default predictability). This approach has also been widely used in prior studies such as Becker and Milbourn (2011) and Badoer and Demiroglu (2018). It is appealing because the fundamental role of credit ratings is to provide insight into future default risk (Becker and Milbourn 2011), and ratings are considered inaccurate if they cannot predict future defaults (Bar-Isaac and Shapiro 2011). Specifically, we estimate the following OLS model[19]:

$$Default_{t+12} = \alpha_0 + \beta_1 Rating_t + \beta_2 Vratio_t + \beta_3 Rating_t \times Vratio_t + \sum_{q=4}^{m} \beta_q (q^{th}\,Control\ Variable_t) + \sum Firm + \sum Year + \varepsilon, \quad (2)$$

where $Default_{t+12}$ is a dummy variable that equals one when there is a default in the next 12 fiscal quarters and zero otherwise. Following Badoer and Demiroglu (2018), we consider that a default occurs if (1) the firm's long-term credit rating is downgraded to C, D, or SD; (2) the firm's outstanding bond issue is assigned a credit rating of C or D by S&P, Moody's, or Fitch; or (3) the firm defaults on a debt obligation or files for bankruptcy. $Rating_t$ and $Vratio_t$ are as defined before. A higher association between $Rating_t$ and $Default_{t+12}$ indicates a given rating's stronger ability to predict future defaults; this translates to higher rating accuracy, which corresponds to a positive coefficient on $Rating_t$ (i.e., $\beta_1 > 0$). If CRAs incorporate unique information from the options market to improve rating accuracy, then the rating-default predictability should become stronger in the presence of more options trading. Thus, the coefficient on the interaction of $Rating_t$ and $Vratio_t$ should be positive (i.e., $\beta_3 > 0$).

In Equations (1) and (2), $Vratio_t$ is lagged relative to the dependent variable to ensure that our results are not driven by changes in options market activity induced by changes in credit rating properties (i.e., reverse causality). The control variables, which are also lagged at time t, include firm characteristics known to influence credit risk. Following Xia (2014), we control for the quarterly leverage (*Leverage*) and return-on-assets (*ROA*), as highly leveraged firms and less-profitable firms face higher credit risk. We also control for asset tangibility (*Tangibility*) because firms with more tangible assets are prone to having higher liquidation values and lower agency costs of debt (Williamson 1988, Harris and Raviv 1990). Because volatile firms are riskier, we include the volatility of leverage (*Lev_Vol*) and the volatility of profitability (*ROA_Vol*). Finally, we include market-to-book (*MTB*) and sales (*Sales*) to account for the possibility that growth potential and sales prospects affect rating decisions. The appendix provides more detailed definitions of all the variables.

We also include firm and year fixed effects in the regressions. To address the concern of potential within-firm correlations, we report t statistics using Huber-White standard errors corrected for firm clustering (Petersen 2009). To ensure that our results are not driven by outliers, we winsorize all continuous variables except for *Rating* at the top and bottom 1%.

3.3. Sample

Because we use two approaches to examine whether CRAs learn from the options market, we construct two separate samples for estimating Equations (1) and (2).[20] For both samples, we begin with all firm-month observations from Compustat with a nonmissing Global Company Key (GVKEY) and long-term credit ratings compiled by S&P from 1996 to 2016. Because we use Option Metrics data to construct all variables related to options trading activities, our sample begins in 1996, the same year that Option Metrics starts its coverage. We then merge these firm-month Compustat observations with Compustat quarterly financial data and keep the data at the firm-quarter level. Next, we eliminate observations with missing options trading data or control variables and observations with ratings worse than "CC." For the analysis of rating-risk responsiveness, we also remove observations with missing *EDP*. The final sample in the analysis of rating-risk responsiveness includes 76,077 firm-quarter observations on 2,599 unique firms, and the final sample in the analysis of rating-default predictability includes 91,961 firm-quarter observations on 3,363 unique firms.

Table 1 presents descriptive statistics for the variables in our baseline analyses. Panel A focuses on the variables used in the analysis of rating-risk responsiveness. The mean $Rating_{t+1}$ is 9.841, corresponding to a BBB− rating, whereas EDP_{t+1} has a mean of 0.050. The mean $Vratio_t$ is 0.039, indicating that the options trading volume is, on average, about 4% of the underlying stock trading volume. Summary statistics for the control variables are comparable to those reported in prior research (Xia 2014). Panel B focuses on the variables used in the analysis of rating-default predictability. The variable $Default_{t+12}$ has a mean value of 0.023, suggesting that 2.3% of the observations in our sample experience at least one default event in the future 12 fiscal quarters, similar to the default rate reported in prior research (Badoer and Demiroglu 2018). $Vratio_t$ is smaller in Panel B than in Panel A, presumably because the larger sample in Panel B includes smaller firms with less options trading activity. The statistics for the variable $Rating_t$ and those of the other control variables are similar to those in Panel A.

4. Main Empirical Results

4.1. Baseline Results

In this section, we report our baseline results on how options trading affects credit rating accuracy. Panel A of Table 2 presents the results from estimating Equation (1), which examines the impact of options trading activity on the responsiveness of ratings to expected credit risk. In column 1, we confirm prior evidence regarding the contemporaneous relation between credit ratings and expected credit risk by estimating Equation (1) without including the interaction between EDP_{t+1} and $Vratio_t$. Consistent with prior research (Kedia et al. 2014, Xia 2014, Bonsall et al. 2015), the coefficient on EDP_{t+1} is positive (1.264) and significant (t statistic = 12.10), suggesting that credit ratings are in line with the credit risk estimates implied by EDP_{t+1}. The coefficient on $Vratio_t$, however, is not significant. The coefficients on the other control variables are consistent with our expectations and those reported in prior research (Xia 2014). For instance, firms with high leverage receive worse ratings, whereas firms with high market-to-book ratios, high asset tangibility, and high sales receive better ratings. In

Table 1. Descriptive Statistics

Panel A: Statistics for the variables used in the analysis of rating-risk responsiveness					
Variable	Mean	Standard deviation	Q1	Median	Q3
$Rating_{t+1}$	9.841	3.382	7.000	10.000	13.000
EDP_{t+1}	0.050	0.163	0.000	0.000	0.001
$Vratio_t$	0.039	0.053	0.006	0.020	0.050
$Leverage_t$	0.322	0.182	0.193	0.299	0.419
ROA_t	0.009	0.023	0.002	0.009	0.019
MTB_t	1.309	0.910	0.772	1.073	1.579
$Tangibility_t$	0.325	0.262	0.098	0.258	0.534
$Sales_t$	6.810	1.418	5.813	6.730	7.767
Lev_Vol_t	0.024	0.028	0.008	0.016	0.029
ROA_Vol_t	0.011	0.017	0.003	0.005	0.011
Panel B: Statistics for the variables used in the analysis of rating-default predictability					
Variable	Mean	Standard deviation	Q1	Median	Q3
$Default_{t+12}$	0.023	0.151	0.000	0.000	0.000
$Rating_t$	10.214	3.461	8.000	10.000	13.000
$Vratio_t$	0.026	0.045	0.000	0.007	0.031
$Leverage_t$	0.344	0.194	0.207	0.319	0.444
ROA_t	0.007	0.047	0.002	0.009	0.018
MTB_t	1.260	0.846	0.764	1.036	1.507
$Tangibility_t$	0.341	0.267	0.107	0.280	0.561
$Sales_t$	6.555	1.509	5.493	6.473	7.562
Lev_Vol_t	0.026	0.038	0.008	0.016	0.030
ROA_Vol_t	0.013	0.032	0.003	0.005	0.012

Notes. This table presents the descriptive statistics for the variables used in the main analyses. Panel A presents descriptive statistics for variables used in the analysis of credit rating responsiveness to expected credit risk, while Panel B presents descriptive statistics for variables used in the analysis of credit ratings' ability to predict future defaults. All variables are defined in the appendix. All continuous variables except *Rating* are winsorized at the top and bottom 1%. The sample in Panel A includes 76,077 firm-quarters for 2,599 unique firms, while the sample in Panel B includes 91,961 firm-quarters for 3,363 unique firms. The sample period is from 1996 to 2016.

Table 2. Effect of Options Trading on the Accuracy of Credit Ratings

Panel A: Rating-risk responsiveness

	$Rating_{t+1}$	
Variable	I	II
EDP_{t+1}	1.264***	1.127***
	(12.10)	(10.12)
$Vratio_t$	0.016	−0.140
	(0.04)	(−0.32)
$EDP_{t+1} \times Vratio_t$		3.752**
		(2.12)
$Leverage_t$	3.009***	3.008***
	(13.77)	(13.77)
ROA_t	−1.903***	−1.891***
	(−3.76)	(−3.75)
MTB_t	−0.227***	−0.227***
	(−5.69)	(−5.69)
$Tangibility_t$	−1.342***	−1.349***
	(−3.93)	(−3.95)
$Sales_t$	−1.029***	−1.029***
	(−16.76)	(−16.78)
Lev_Vol_t	0.843**	0.853**
	(2.42)	(2.45)
ROA_Vol_t	6.082***	6.107***
	(8.33)	(8.36)
Constant	17.009***	17.019***
	(35.57)	(35.63)
Adjusted R^2	0.900	0.900
Observations	76,077	76,077

Panel B: Rating-default predictability

	$Default_{t+12}$	
Variable	I	II
$Rating_t$	0.007***	0.006***
	(4.08)	(3.37)
$Vratio_t$	0.086	−0.388***
	(1.62)	(−3.20)
$Rating_t \times Vratio_t$		0.050***
		(2.98)
$Leverage_t$	0.032	0.035
	(1.49)	(1.62)
ROA_t	−0.096***	−0.096***
	(−3.55)	(−3.57)
MTB_t	−0.008***	−0.008***
	(−3.84)	(−3.91)
$Tangibility_t$	0.045	0.045
	(1.40)	(1.41)
$Sales_t$	0.006	0.006
	(1.32)	(1.34)
Lev_Vol_t	−0.017	−0.016
	(−0.55)	(−0.53)
ROA_Vol_t	−0.045	−0.042
	(−0.90)	(−0.83)
Constant	−0.140***	−0.127***
	(−3.25)	(−3.01)
Adjusted R^2	0.305	0.307
Observations	91,961	91,961

Notes. This table presents the regression results for the relation between options trading and credit rating accuracy. Panel A provides rating accuracy using the responsiveness of ratings to expected credit risk, while Panel B presents rating accuracy using the ability of credit ratings to predict future default. The sample period is from 1996 to 2016. All variables are defined in the appendix. All continuous variables except *Rating* are winsorized at the top and bottom 1%. We include year and firm fixed effects in our regression analyses. We estimate robust standard errors clustered by firm. The *t* statistics are presented in parentheses.

*, **, and ***Statistical significance at the 10%, 5%, and 1% levels, using two-tailed tests, respectively.

column 2, we add the interaction of EDP_{t+1} with $Vratio_t$. We find a positive and significant coefficient on $EDP_{t+1} \times Vratio_t$ (coefficient = 3.752, t statistic = 2.12), suggesting that the contemporaneous responsiveness of credit ratings to expected credit risk increases in lagged options trading volume. The coefficients on other variables are similar to those in column 1.

Panel B of Table 2 presents the results from estimating Equation (2), which examines how options trading activity is related to the ability of credit ratings to predict future defaults. Column 1 presents the results without the interaction between $Rating_t$ and $Vratio_t$. Consistent with expectations, the coefficient on $Rating_t$ is positive and significant (coefficient = 0.007, t statistic = 4.08), suggesting that worse ratings (i.e., higher numerical values of *Rating*) correspond to a higher likelihood of future defaults. Furthermore, the coefficient on $Vratio_t$ is insignificant, whereas the coefficients on other variables are largely in line with expectations. For example, firms with high profitability and high market-to-book ratios are less likely to default in the future. In column 2, we add the interaction of $Rating_t$ with $Vratio_t$. The coefficient on this interaction term is significantly positive (coefficient = 0.050, t statistic = 2.98), indicating that ratings exhibit a stronger ability to predict future defaults when there is a higher level of options trading activity. The coefficients on the remaining variables are generally similar to those in column 1.

In summary, the results in Table 2 show that credit ratings become more responsive to expected credit risk and exhibit higher ability to predict future defaults when options trading activity is greater. The collective evidence is consistent with rating agencies gleaning new information from the options market, which, in turn, increases the accuracy of their ratings.

4.2. Endogeneity Issues

Our causal interpretation of the positive association between options trading activity and credit rating accuracy could be problematic if unidentified factors simultaneously drive options trading activity and rating accuracy. Moreover, changes in options market activity might be induced by changes in credit rating properties (i.e., reverse causality). Our baseline empirical specifications should alleviate these endogeneity concerns, as (1) we control for several firm characteristics that prior studies identify as determinants of credit risk (e.g., leverage, profit volatility), (2) we lag the options trading variable relative to the rating accuracy variable, and (3) we include firm fixed effects in our analyses, which should mitigate endogeneity bias that stems from time-invariant firm characteristics. To further address endogeneity concerns and corroborate our causal interpretation, we perform a difference-in-differences analysis discussed later.[21]

We conduct our difference-in-differences analysis using a sample of firms with options listed for the first time and a matched control sample without listed options over the same period. Similar to Naiker et al. (2013), we use this approach to examine the change in rating accuracy one year after option listing versus one year prior for firms with option listing relative to firms without a listed option during the same period. Option listing decisions are made by exchanges and are plausibly exogenous to firm decisions; however, option listings can still be endogenous if the options exchanges selectively list firms for options trading based on certain firm characteristics (Mayhew and Mihov 2004). In practice, options exchanges identify firms for option listing based on the eligibility criteria that are set by the SEC; hence, it is unlikely that options exchanges selectively list firms.[22] Nonetheless, following Mayhew and Mihov (2004), we construct a matched sample based on ex ante probabilities of option listing.

Specifically, we identify treatment firms with new option listing during our sample period. Then, we select the control firms based on ex ante probabilities of option listing from firms that have no listed options during the same period but have all the control variables used in our main analyses. We estimate the ex ante probabilities of option listing using a probit model with the following explanatory variables: average daily stock trading volume over the 250 trading days prior to option listing date (*S_Volume*), the annualized standard deviation of log stock return over the same period (*STD*), the ratio of 30-day to 250-day average daily trading volume prior to option listing date (*ABVOL*), the ratio of 30-day to 250-day annualized standard deviation of log stock return prior to option listing date (*ABSTD*), firm size (*LMV*), all control variables in our baseline models Equation (1) and (2), and year and firm fixed effects. We then construct the control sample by using the nearest neighbor method to select the stocks with propensity scores closest to those of the treatment firms within the same (SIC two-digit) industry.[23]

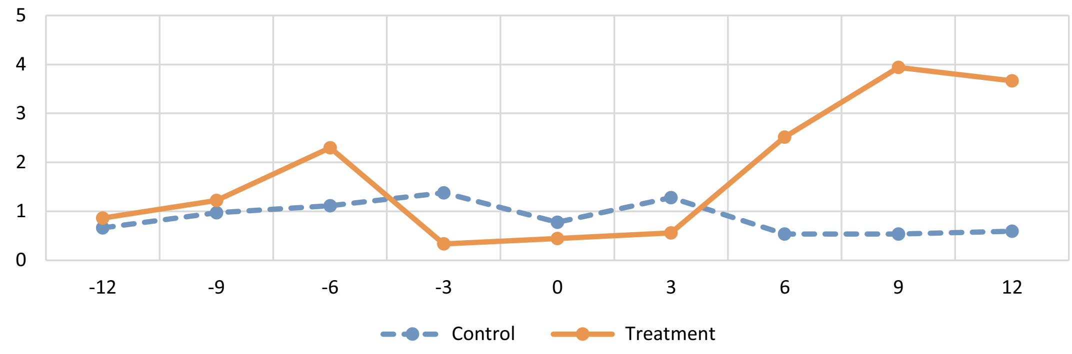
Using this matched sample, we first test whether the parallel-trends assumption is satisfied by regressing *Rating* on the concurrent *EDP* for each quarter over the eight-quarter period around the option-listing event and then plotting the coefficients in Figure 1. This figure shows that rating-risk responsiveness is similar in the prelisting period between firms that subsequently have listed options and firms that do not have listed options. In contrast, credit ratings become more responsive to expected credit risk for listed firms after option listing than for nonlisted firms.

Next, we estimate the following OLS model using our matched sample:

$$\begin{aligned} Rating_{t+1} = {} & \alpha_0 + \beta_1 EDP_{t+1} + \beta_2 Post_t + \beta_3 List_t + \beta_4 Post_t \\ & \times List_t + \beta_5 EDP_{t+1} \times Post_t + \beta_6 EDP_{t+1} \times List_t \\ & + \beta_7 EDP_{t+1} \times List_t \times Post_t \\ & + \sum_{q=8}^{m} \beta_q (q^{th} Control\ Variable_t) + \varepsilon, \qquad (3) \end{aligned}$$

where *Post* is an indicator variable that equals one for observations within one year after the first month with nonzero options trading volume, and zero for observations one year prior to the first nonzero options trading month.[24] *List* is an indicator that equals one for firms with options listed (treatment firms), and zero for firms without listed options (control firms). We identify the occurrence of an options-listing event based on the first day when the options exchange reports a nonzero options trading volume. For explanatory variables, we adopt best practices recommended by Shipman et al. (2017) and include all control variables in our baseline models and the additional explanatory variables from the first stage model used to predict option listing probabilities (i.e., *S_Volume*, *STD*, *ABVOL*, *ABSTD*, *LMV*), as

Figure 1. (Color online) Rating-Risk Responsiveness Before and After Option Listing



Notes. This figure presents the time series variation of rating-risk responsiveness, that is, the regression coefficient on *EDP* from regressing *Rating* on the concurrent *EDP* each quarter over the eight-quarter period around the option-listing event. The solid line represents the treatment firms for which there is an option-listing event. The dotted line represents the control firms for which there is no option-listing event during the same period.

well as year and firm fixed effects. The main variable of interest is $EDP_{t+1} \times List_t \times Post_t$. Given our hypothesis that rating accuracy should increase in options trading activity, we expect the coefficient on this interaction term to be positive.

The results of these analyses are reported in Table 3. Columns I and II present the results without and with the control variables, respectively. In both columns, the coefficient on EDP_{t+1} is positive and significant, indicating that credit ratings, on average, are consistent with credit risk estimates implied by EDP_{t+1}. These two columns also show that the coefficient on the three-way interaction term $EDP_{t+1} \times List_t \times Post_t$ is significantly positive, suggesting that the mapping between expected default probabilities and credit ratings becomes stronger following the option listing event for listed firms relative to nonlisted firms over the same period.

To further check the robustness of our results, we repeat the above analyses over a shorter horizon and examine the change in rating accuracy over the six-month period after the option listing event relative to the six-month period before. As shown in columns III and IV of Table 3, the coefficients on $EDP_{t+1} \times List_t \times Post_t$ remain significantly positive. In addition, the coefficient on $EDP_{t+1} \times List_t$ is insignificant across all four columns, suggesting that rating-risk responsiveness is not significantly different in the prelisting period between firms that subsequently have listed options and firms that do not have listed options. This result, combined with the visual evidence in Figure 1, suggests that the parallel-trends assumption is satisfied and confirms the validity of the difference-in-differences analysis. In short, Table 3 provides further support that rating accuracy improves in response to increased options trading volume.[25]

5. Underlying Mechanisms

In this section, we seek to corroborate our inference that CRA learning is what drives the positive association between options trading and credit rating accuracy documented in the previous section. We use two approaches to shed light on the underlying mechanisms. The first approach is to test whether ratings are more sensitive to the options market inputs in the presence of higher options trading volume. The second approach is to conduct cross-sectional analyses based on when CRAs are more likely to seek additional information from the options market.[26]

5.1. Options Trading and the Sensitivity of Credit Ratings to Information in Option Prices

In this section, we examine the relation between options trading activity and the sensitivity of credit ratings to ratings-relevant information that can be extracted

Table 3. Difference-in-Differences Analysis Based on Option Listing for Rating-Risk Responsiveness

	Estimation window			
	$Rating_{t+1}$			
Variable	I 1 Year	II 1 Year	III 6 months	IV 6 months
EDP_{t+1}	1.322***	0.613***	0.839***	0.570*
	(7.85)	(3.71)	(2.68)	(1.86)
$Post_t$	0.0901**	0.0621*	0.0823*	0.0680
	(2.37)	(1.71)	(1.94)	(1.64)
$List_t$	−0.186***	−0.119***	−0.146***	−0.0847
	(−3.99)	(−2.65)	(−2.75)	(−1.61)
$Post_t \times List_t$	0.0671	0.0119	−0.0202	−0.0517
	(1.46)	(0.27)	(−0.38)	(−1.00)
$EDP_{t+1} \times Post_t$	−0.653***	−0.647***	−0.296	−0.184
	(−2.99)	(−3.09)	(−0.82)	(−0.52)
$EDP_{t+1} \times List_t$	−0.0504	0.0112	0.580	0.513
	(−0.24)	(0.06)	(1.59)	(1.44)
$EDP_{t+1} \times List_t \times Post_t$	1.318***	1.141***	0.919**	0.785*
	(4.51)	(4.07)	(2.00)	(1.75)
Controls	No	Yes	No	Yes
Adjusted R^2	0.941	0.946	0.949	0.952
Observations	6,345	6,345	3,853	3,853

Notes. This table presents the difference-in-difference analysis based on option listing for the credit ratings' responsiveness to expected credit risk. The sample is constructed using propensity score matching based on the following variables that are related to the option listing decision: stock trading volume (*S_Volume*), stock return standard deviation (*STD*), abnormal stock trading volume (*ABVOL*), abnormal stock return standard deviation (*ABSTD*), firm size (*LMV*), all control variables in our baseline models, and year and firm fixed effects. All variables are defined in the appendix. All continuous variables, except *Rating*, are winsorized at the top and bottom 1%. We include year and firm fixed effects in our regression analysis. The *t* statistics are presented in parentheses.

*, **, and ***Statistical significance at the 10%, 5%, and 1% levels, using two-tailed tests, respectively.

exclusively from option prices. As discussed in Section 2, option prices reveal nondirectional information about future stock return volatility and directional information about future negative news, both of which can be particularly useful for CRAs. Using implied volatility skews to forecast future volatility and performance, Xing et al. (2010) show that informed traders with negative news prefer to trade out-of-the-money put options, which is precisely the type of nonpublic information that CRAs would want as quickly as possible. Similarly, Jin et al. (2012) use implied volatility skews and spreads to show that option prices/trades display significantly greater predictive abilities around corporate announcements than do stock prices/trades.

If CRAs extract information from the options market to assess credit risk, then we expect higher options trading volume to result in greater rating sensitivity to measures of implied volatility derived from option prices. To test this conjecture, we construct three measures of implied volatility using option prices: implied volatility levels (*IV_ATM*), skews (*Skew*), and spreads (*Spread*). We then modify Equation (1) by substituting EDP_{t+1} with IV_ATM_t, $Skew_t$, and $Spread_t$ one at a time. Our CRA learning hypothesis predicts that the sensitivity of credit ratings to each measure of implied volatility should become stronger as options trading volume increases.

Table 4 reports the results. Column I uses IV_ATM_t, whereas columns II and III use $Skew_t$ and $Spread_t$, respectively. In column I, the coefficient on IV_ATM_t is significantly positive, indicating that a higher implied volatility level is associated with worse ratings. More importantly, the coefficient on the interaction of IV_ATM_t and $Vratio_t$ is significantly positive, suggesting that the relation between implied volatility level and credit ratings becomes stronger as options trading volume increases. In columns II and III, the coefficients on the interactions of $Vratio_t$ with both $Skew_t$ and $Spread_t$ are significantly positive, suggesting that as options trading volume increases, credit ratings become more sensitive to implied volatility skews and spreads.[27]

Although the results in columns I, II, and III confirm that CRAs use the options market as a direct source of information in their ratings, it is also possible that CRAs learn indirectly from the options market through its impact on other markets (i.e., stock and bond markets) as discussed in Section 2. To examine more closely the effects of direct and indirect learning, we add stock return volatility (Ret_STD_t), bond yield spread (YS_t), and their interactions with $Vratio_t$ into the models in

Table 4. Options Trading and Credit Rating Sensitivity to Information in Option Prices

	$Rating_{t+1}$					
Variable	I	II	III	IV	V	VI
$Vratio_t$	−2.301**	−1.340**	−0.380	−1.635*	−1.786**	−0.322
	(−2.54)	(−2.02)	(−0.77)	(−1.82)	(−2.18)	(−0.49)
IV_ATM_t	0.254***			0.261***		
	(17.69)			(9.89)		
$IV\_ATM_t \times Vratio_t$	0.522**			0.656*		
	(2.44)			(1.96)		
Skewt		1.876***			0.772	
		(4.16)			(1.31)	
$Skew_t \times Vratio_t$		22.749***			18.859**	
		(2.80)			(2.48)	
$Spread_t$			0.185***			0.095
			(3.86)			(1.40)
$Spread_t \times Vratio_t$			2.522***			2.323***
			(3.61)			(3.32)
Ret_STD_t				−11.822***	7.859***	10.708***
				(−5.37)	(4.90)	(7.16)
$Ret\_STD_t \times Vratio_t$				−18.821	64.338***	32.276
				(−0.53)	(2.71)	(1.55)
YS_t				4.440***	5.459***	5.185***
				(3.60)	(3.84)	(4.03)
$YS_t \times Vratio_t$				40.718	49.784	49.655
				(0.98)	(1.13)	(1.20)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.902	0.894	0.895	0.912	0.908	0.910
Observations	64,092	59,581	63,737	33,278	31,651	33,223

Notes. This table presents the results on the relation between options trading and the sensitivity of credit ratings to the information in option prices (i.e., implied volatility level, skew, and spread). The sample period is from 1996 to 2016. All variables are defined in the appendix. All continuous variables except *Rating* are winsorized at the top and bottom 1%. We include year and firm fixed effects in our regression analyses. We estimate robust standard errors clustered by firm. The *t* statistics are presented in parentheses.

*, **, and ***Statistical significance at the 10%, 5%, and 1% levels, using two-tailed tests, respectively.

columns I, II, and III. As shown in columns IV, V, and VI, the coefficients on the interactions of $Vratio_t$ with IV_ATM_t, $Skew_t$, and $Spread_t$ remain positive and significant. In addition, we find mostly positive coefficients on the interactions of $Vratio_t$ with Ret_STD_t and YS_t, though not always significant, suggesting that CRAs also learn indirectly through spillover effects from the options market to the stock and bond markets.[28] Collectively, the results in Table 4 confirm that CRAs extract ratings-relevant information directly from the options market as trading volume increases. We also find some evidence consistent with the notion of indirect learning.

5.2. Cross-Sectional Analyses

Next, we examine cross-sectional variations in the relation between options trading activity and rating accuracy. We consider two key factors that could influence CRAs' tendency to seek additional sources of information to enhance the accuracy of their ratings: (1) firm default risk and (2) informativeness of options trading.

5.2.1. Firm Default Risk and CRA Learning. When the firm is closer to default, any additional risk will have a higher incremental effect on default probabilities (Pan et al. 2018). Thus, when rating firms that have a high default risk, CRAs are more likely to seek additional information from other sources because failure to predict defaults may damage CRAs' reputation. In particular, CRAs are likely to turn to the options market for additional information, because option prices rapidly reveal both directional information about future negative news and nondirectional information about future volatility (Ni et al. 2008, Xing et al. 2010). Furthermore, creditors of firms that are close to default may pressure CRAs to reassess the credit worthiness of the firm in light of the credit risk related information from the options market. Timely access to information about the firm's credit risk is critical for these creditors, as each creditor may have an incentive to be the first to force a liquidation of the firm's assets to guarantee full payment (Hotchkiss et al. 2008). We therefore posit that the impact of options trading on rating accuracy will be more pronounced among firms with higher default risk.

To test this prediction, we capture firm default risk via the Z-score of Altman (1968) (*Altman Z*) and leverage (*Leverage*). Higher values of *Altman Z* imply a lower probability of bankruptcy and hence lower default risk, whereas higher values of *Leverage* imply a higher default risk. We then estimate Equations (1) and (2) after partitioning the corresponding samples into high default risk versus low default risk groups based on the sample terciles of *Altman Z* and *Leverage*.

Table 5 presents the results of these analyses. Panel A provides the results when we use rating-risk responsiveness to capture rating accuracy and shows that options trading has a significant effect on rating-risk responsiveness only among firms with high default risk, identified by low *Altman Z* (column I) and high *Leverage* (column IV). Panel B presents the results when we use rating-default predictability to capture rating accuracy and shows similar results when we use *Altman Z* and *Leverage* to capture default risk. Overall, the results in Table 5 are in line with the idea that CRAs rely more on option markets for information when assessing firms with high default risk.

Table 5. Firm Default Risk and CRA Learning

Panel A: Rating Risk responsiveness

	Altman Z		*Leverage*	
	I	II	III	IV
Variable	*T1 (Low)* $Rating_{t+1}$	*T3(High)* $Rating_{t+1}$	*T1(Low)* $Rating_{t+1}$	*T3(High)* $Rating_{t+1}$
EDP_{t+1}	1.124***	1.312***	0.941**	0.647***
	(5.97)	(7.56)	(2.36)	(5.44)
$Vratio_t$	−0.472	0.005	−0.696	1.332*
	(−0.57)	(0.01)	(−1.03)	(1.82)
$EDP_{t+1} \times Vratio_t$	8.021**	1.224	−3.653	3.313*
	(2.33)	(0.48)	(−0.55)	(1.85)
Controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.910	0.907	0.925	0.895
Observations	25,332	25,358	24,998	25,029

Panel B: Rating-default predictability

	Altman Z		*Leverage*	
	I	II	III	IV
Variable	*T1 (Low)* $Default_{t+12}$	*T3(High)* $Default_{t+12}$	*T1(Low)* $Default_{t+12}$	*T3(High)* $Default_{t+12}$
$Rating_t$	0.009**	0.005*	0.007***	0.010***
	(2.29)	(1.80)	(2.60)	(2.62)
$Vratio_t$	−0.846***	0.064	−0.028	−0.756***
	(−3.37)	(1.24)	(−0.17)	(−2.90)
$Rating_t \times Vratio_t$	0.111***	−0.010	0.001	0.076***
	(3.44)	(−1.24)	(0.06)	(2.71)
Controls	Yes	Yes	Yes	Yes
Observations	30,610	30,641	30,627	30,654
Adjusted R^2	0.386	0.428	0.473	0.392

Notes. This table presents the regression results of the cross-sectional analysis of the relation between options trading and credit rating accuracy based on firm default risk. Panel A provides rating accuracy using rating responsiveness to expected credit risk, while Panel B presents rating accuracy using rating default predictability. *T1* refers to the bottom tercile of the partitioning variable, whereas *T3* refers to the top tercile. All other variables are defined in the appendix. All continuous variables, except *Rating*, are winsorized at the top and bottom 1%. Year and firm fixed effects are included. We estimate robust standard errors clustered by firm. The *t* statistics are presented in parentheses.

*, **, and ***Statistical significance at the 10%, 5%, and 1% levels, using two-tailed tests, respectively.

5.2.2. Informativeness of Options Trading and CRA Learning. The amount of new information impounded into option prices depends on the extent to which informed traders choose to transact in options instead of other securities (e.g., stocks, bonds). When more informed traders choose to transact in options, options trading becomes more informative and CRAs are more able to learn from the options market and improve their rating accuracy. Thus, we expect that the impact of options trading on credit rating accuracy will be stronger when options trading is more informative.

To test this prediction, we capture informativeness of option trading using the ratio of trading volume in out-of-the-money options relative to that in all traded options *(Out of the Money)*, and the ratio of trading volume in short-maturity options relative to that in all traded options *(Short OM)*. Out-of-the-money options and short maturity options are both attractive to informed traders because the former provide traders with greater leverage (Pan and Poteshman 2006), and the latter are more liquid (Pan and Poteshman 2006, Roll et al. 2009). We then estimate Equations (1) and (2) by partitioning the corresponding samples into high-informativeness versus low-informativeness groups, based on the sample terciles of *Out of the Money* and *Short OM.*

The results are reported in Table 6. Panel A presents the results when we use rating-risk responsiveness to capture rating accuracy and shows that options trading has a significantly positive effect on rating-risk responsiveness only for firms in the top tercile of *Out of the Money* (*Short OM*). In Panel B, where we use rating-default predictability as the measure of rating accuracy, we similarly find that the effect of options trading on rating-default predictability is positive and significant only for firms in the top tercile of *Out of the Money* (*Short OM*). These results suggest that when options trading is more informative, CRAs learn more from the options market, which, in turn, further improves ratings accuracy.[29]

Table 6. Informativeness of Options Trading and CRA Learning

Panel A: Rating-risk responsiveness

	Out of the money		*Short OM*	
	I	II	III	IV
Variable	*T1 (low)* $Rating_{t+1}$	*T3 (high)* $Rating_{t+1}$	*T1 (low)* $Rating_{t+1}$	*T3 (high)* $Rating_{t+1}$
EDP_{t+1}	1.157***	0.784***	1.375***	0.679***
	(8.03)	(4.08)	(8.92)	(4.15)
$Vratio_t$	0.615	−0.060	−0.512	0.020
	(0.56)	(−0.14)	(−0.43)	(0.05)
$EDP_{t+1} \times Vratio_t$	1.450	6.681***	−2.138	7.412***
	(0.30)	(3.08)	(−0.56)	(3.04)
Controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.913	0.908	0.901	0.911
Observations	24,402	24,487	25,329	25,362

Panel B: Rating-default predictability

	Out of the money		*Short OM*	
	I	II	III	IV
Variable	*T1 (low)* $Default_{t+12}$	*T3 (high)* $Default_{t+12}$	*T1 (low)* $Default_{t+12}$	*T3 (high)* $Default_{t+12}$
$Rating_t$	0.009***	0.004	0.008*	0.003
	(2.96)	(1.60)	(1.77)	(1.21)
$Vratio_t$	−0.638*	−0.492***	0.175	−0.431**
	(−1.85)	(−2.81)	(0.49)	(−2.24)
$Rating_t \times Vratio_t$	0.071	0.065***	−0.015	0.057**
	(1.43)	(2.88)	(−0.30)	(2.41)
Controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.413	0.361	0.461	0.369
Observations	23,523	23,596	25,838	23,977

Notes. This table presents the regression results of the cross-sectional analysis of the relation between options trading and credit rating accuracy based on informativeness of options trading. Panel A provides rating accuracy using rating responsiveness to expected credit risk, while Panel B presents rating accuracy using rating default predictability. *T1* refers to the bottom tercile of the partitioning variable, while *T3* refers to the top tercile. All other variables are defined in the appendix. All continuous variables, except *Rating*, are winsorized at the top and bottom 1%. Year and firm fixed effects are included. We estimate robust standard errors clustered by firm. The *t* statistics are presented in parentheses.

*, **, and ***Statistical significance at the 10%, 5%, and 1% levels, using two-tailed tests, respectively.

6. Conclusion

Do CRAs learn new information from the options market and use this information to improve credit rating accuracy? Our empirical findings show that as options trading volume increases, credit ratings become more responsive to expected credit risk and exhibit greater ability to predict future defaults. These results are robust to a number of sensitivity checks and additional tests, including difference-in-differences analyses in which the options listing serves as an exogenous shock to options trading activities. The evidence consistently shows that CRAs incorporate unique information from the options market into their rating decisions, which leads to improved credit rating accuracy.

We use two approaches to examine our posited learning mechanism that underlies the positive relation between options trading and rating accuracy. The first approach examines the relation between options trading volume and credit rating sensitivity to implied volatility measures estimated exclusively from options prices. We find that when options trading volume increases, credit ratings become more sensitive to implied volatility levels, skews, and spreads. Furthermore, in addition to CRAs learning directly from the options market, we find some evidence that CRAs learn indirectly through information spillovers from the options market into other markets. The second approach uses cross-sectional analyses to examine conditions under which CRAs are more likely to seek information from the options market. We find that the impact of options trading on credit rating accuracy is stronger when firm default risk is higher, options trading is more informative, manager-provided information is of lower quality, and firm uncertainty is higher. These results provide further support for the learning mechanism.

Overall, this study provides new evidence that CRAs incorporate information from the options market into their credit rating decisions to improve rating accuracy. This is an important finding because, unlike other market participants, CRAs have privileged access to private information through their corporate client relationships (Bonsall 2014). Our results show that despite CRAs' privileged access to private information from client management, the options market still provides valuable information to these information intermediaries.

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Appendix. Definition of Variables

Variable	Definition
Rating	The assigned numerical monthly long-term issuer's credit ratings issued by S&P averaged over the fiscal quarter. The alphabetical rating system is converted to a cardinal system in which an "AAA" rating equals 1, an "AA" rating equals 2, etc. Hence, the highest numerical rating of 20 is assigned to the "CC" rating. We do not include "C", "D," or "SD" ratings in our sample because these ratings are indicator of defaults or issued after defaults.
Default	A dummy variable that takes the value one if the firm experiences a default in the future 12 fiscal quarters, where default is defined as (1) the long-term issuer's credit rating is downgraded to C, D, or SD; (2) the firm's outstanding debt is issued a credit rating of C or D by S&P, Moody's, or Fitch; or (3) the firm defaults on a debt obligation or files for bankruptcy. We collect the data on default, bankruptcy, and bond issue ratings from the Mergent Fixed Income Securities Database (FISD).
EDP	The firm's daily expected default probability derived based on the Merton (1974)/KMV model averaged over the fiscal quarter.
Vratio	The ratio of the quarterly options trading volume to the underlying stock's trading volume over the same fiscal quarter. Because one exchange-traded option contract represents 100 shares of the underlying stock, we multiply this ratio by 100 to make this ratio easier to interpret.
Leverage	The quarterly leverage ratio calculated as the total long-term and current liabilities deflated by the total assets.
ROA	The quarterly return on assets defined as income before extraordinary items scaled by the average total assets.
MTB	The quarterly market-to-book ratio measured as the market value of total equity divided by the book value of total equity.
Tangibility	The quarterly asset tangibility defined as the net property, plant, and equipment deflated by total assets.
Sales	The natural logarithm of total quarterly sales.
Lev_Vol	The standard deviation of *Leverage* estimated over the past four fiscal quarters.
ROA_Vol	The standard deviation of *ROA* estimated over the past four fiscal quarters.
IV_ATM	The implied volatility of at-the-money call options averaged over the fiscal quarter.
Skew	The daily *Skew* averaged over the fiscal quarter. The daily *Skew* is the difference between the out-of-the-money put option's weighted average implied volatility and the at-the-money call option's weighted average implied volatility, where the weight is each individual option contract's daily open interest.
Spread	The weighted average difference between implied volatilities of put and call options matched on strike price and maturity dates, where the weight is the combined daily open interest of call and put option pair.
Earnings Vol	The standard deviation of quarterly income before extraordinary items (scaled by average total assets) in the past 12 fiscal quarters.
AF Dispersion	The standard deviation of analyst earnings per share (EPS) forecasts for the current quarter scaled by the actual EPS for the same quarter.
Meet or Beat	A dummy variable that equals one if the firm's quarterly net income scaled by total assets is greater than or equal to zero but less than 0.005.
Insider CAR	The average profitability of insiders' trades over the fiscal quarter. Trade profitability is equal to the one-month market adjusted return following the trade by the insider. Insider trades include any open market stock transaction initiated by the top five executives from Thomson Insider Data.
Altman Z	Altman's Z-score calculated as: $$Z = 0.012 * X_1 + 0.014 * X_2 + 0.033 * X_3 + 0.006 * X_4 + 0.999 * X_5$$ where X_1 = Working Capital/Total Assets; X_2 = Retained Earnings/Total Assets; X_3 = Earnings before Interest and Taxes/Total Assets; X_4 = Market Value of Equity/Book Value of Total Liabilities; X_5 = Sales/Total Assets; Z = Overall Index.
Out of the Money	The percentage of option volume that is out of the money, where we define out of the money options as call options that have a delta between 0.125 and 0.375 and put options that have a delta between −0.375 and −0.125.
Short OM	The percentage of options volume that is attributable to option contracts that expire within two months.
LMV	Natural logarithm of market value of equity.
LNA	Natural logarithm of 1 plus the total number of analysts who cover the firm.
InstOwn	Percentage of outstanding shares owned by institutions.

Appendix. (Continued)

Variable	Definition
Post	A dummy variable that equals one for the firm-quarter observations in the period after option listing, and zero otherwise, where option listing date is defined as the first day with a none-zero options trading volume.
List	A dummy variable that equals one for firm-quarter observations that have option listing events during the sample period, and zero otherwise.
Open_Interest	The total number of daily call and put option contracts that have not been settled over the fiscal quarter.
Investment	The quarterly capital expenditures scaled by total assets at the beginning of the quarter.
DS	Debt structure calculated as the ratio of the outstanding amount of public debt from FISD to total debt as of the end of the fiscal quarter.
Payout	Dividend payout calculated as the ratio of cash dividends to stock price.
Guide	A dummy variable that equals one when the firm issues management guidance during the fiscal quarter, and zero otherwise.
Ret_STD	The standard deviation of stock return during the fiscal quarter.
YS	The weighted average yield spread of outstanding bond issues, where the weight is the outstanding amount of each issue.
Ret	The stock return during the fiscal quarter.
OV	The total value of options trading volume during the fiscal quarter, expressed in millions of underlying equity shares.
LOV	Natural logarithm of total options trading volume over the fiscal quarter.
LOMV	Natural logarithm of the dollar value of all option contracts traded for a firm during the fiscal quarter.
OSMVR	The ratio of options trading volume to stock trading volume (in dollars) over the fiscal quarter.
ORQS	Ordinal rating quality shortfall measured as the absolute difference between the quarterly rank of credit rating and the quarterly rank of expected default probability normalized by the sample size. *ORQS* ranges between 0 and 1, where 0 represents the most accurate rating.

Endnotes

[1] See http://www.cboe.com/data/historical-options-data/volume-put-call-ratios.

[2] See https://www.cnbc.com/2020/07/24/options-volume-hits-record-with-bullish-sentiment-at-extremely-high-levels-goldman-sachs-says.html.

[3] Nonpublic information refers to information that has not been disseminated to the general public. Such information could be private, or insider information (e.g., upcoming merger announcements or new product launches), or proprietary information that was uniquely processed from public sources (e.g., stock value forecasts from in-house pricing models or bond value forecasts from in-house default prediction models).

[4] Blanco and Wehrheim (2017) and Roll et al. (2009) show that the informational benefits of the options market are conditional on the volume of options trading. Another attractive feature of using option trading volumes (rather than prices) is that, if the options market is preferred by informed traders, then changes in option trades will reflect new information before changes in stock prices or option prices (Easley et al. 1998).

[5] Although actual defaults are rare, examining default probabilities is still an appealing approach to capture rating accuracy. Bonsall et al. (2017) state that (p. 55), "we view the use of actual defaults as the strongest of tests of rating accuracy because assessing the relative likelihood of actual default events is the main objective of CRAs."

[6] One possible alternative explanation for our results is that firm managers learn new information from the options market and subsequently make firm decisions based on this information, which then affect credit rating accuracy. Prior work shows that options trading activities alter several managerial behaviors such as investment (Blanco and Wehrheim 2017, Roll et al. 2010), debt structure (Cao et al. 2023), and voluntary disclosure (Chen et al. 2021). To rule out this explanation, we control for the effects of several firm decisions, including investment, debt structure, dividend payout, and voluntary disclosure and continue to find similar results (Table OA5 in the online appendix).

[7] For brevity, we discuss these robustness checks in the online appendix (Table OA8). To further check the robustness of our results, we also assess rating accuracy using the ability of ratings to explain expected default risk as measured by the failure score of Campbell et al. (2008). We find that ratings have a greater ability to explain expected default risk as option trading volume increases (untabulated), which again supports our hypothesis. All untabulated results are available upon request.

[8] For this reason, we test the sensitivity of our baseline results to potential correlated omitted variables using the impact threshold of a confounding variable (ITCV), as recommended by Larcker and Rusticus (2010) and Frank (2000). We find that a confounding variable would have to be at least 1.22 (3.24) times larger than the most impactful control variable to overturn our baseline results for rating-risk responsiveness (rating-default predictability) (Table OA4 of the online appendix).

[9] We discuss the cross-sectional analyses based on quality of manager-provided information and firm uncertainty in the online appendix (Tables OA6 and OA7, respectively).

[10] See Mayhew and Mihov (2004) for additional details on initial option listing requirements.

[11] Although our study is related to Cremers et al. (2008) in some aspects, the two studies are quite different in terms of the fundamental research question, data, methodology, findings, and implications. First, Cremers et al. (2008) focus on the determinants of *credit risk* as reflected in credit spreads, while our study focuses on the determinants of *credit rating accuracy* as captured by credit ratings' responsiveness to expected credit risk and ability to predict future defaults. Empirically, Cremers et al. (2008) use credit spreads of *corporate bonds* from the Bloomberg Corporate Bonds database, whereas we use S&P long-term issuer ratings from Compustat to evaluate the accuracy of *firm* ratings, rather than individual bond ratings. Second, although both papers find that the options market serves a useful informational role, the two studies differ with respect to the end users (bond investors versus CRAs), methods, and purposes of such information

extraction. Third, the economic implications of the two studies are dissimilar. Cremers et al. (2008) suggests that bond pricing models can be improved by incorporating implied volatilities/skews as explanatory variables. Our study shows that information intermediaries such as CRAs can improve the quality of their information provision services by extracting incremental information from the options markets.

[12] Previous studies document that reputation is a first-order consideration for CRAs and has a significant impact on their future economic rents (House 1995, Meltz 2007, Pitman 2008).

[13] We conduct six interviews with current and former managers and analysts from the three major CRAs (i.e., S&P Global Ratings, Moody's Investor Services, and Fitch Ratings). These interviews confirm that CRAs use information from capital markets, including the options market, in their rating process. Rating analysts regularly monitor the capital markets for potential signals that might be relevant for credit rating re-evaluation. If they notice divergence between the affected firm's current ratings and potentially-revised ratings based on capital market signals, they conduct further investigations to explore whether and how credit ratings should be adjusted. In particular, one rating manager mentioned that rating analysts look at number of put options when assessing credit risk.

[14] Rating agencies were exempt from Regulation Fair Disclosure (Reg FD) from October 2000 until the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 removed their exemption; however, issuer-pay rating agencies continue to have access to private information by entering into confidentiality agreements with issuers (Gibson, Dunn & Crutcher LLP 2010, Quinlivan 2010).

[15] Similarly, the literature on the feedback effects of financial markets on managerial decisions emphasizes that, even if an individual investor has less information than the manager, the market could still be more informed in some way, as it aggregates pieces of information from a large number of investors (Grossman 1976, Hellwig 1980, Bond et al. 2012, Zuo 2016).

[16] That limited resources could reduce the capacity of information intermediaries to collect, process, and disseminate information has been shown in various contexts. For example, D'Souza et al. (2010) show that S&P, which disseminates corporate accounting information via Compustat, exhibits higher information collection lags during peak market activity (e.g., high 10-K report filing intensity), consistent with S&P's having limited resources and capacity. In addition, Akbas et al. (2018) show that Thomson Reuters, another information intermediary, faces resource constraints in processing and disseminating analyst forecasts via I/B/E/S.

[17] Another attractive feature of using options trading volume (rather than prices) is that, if the options market is preferred by informed traders, then changes in option trades will reflect new information before changes in option prices or stock prices (Easley et al. 1998).

[18] Specifically, we follow prior studies and assign numerical values to each letter rating on a notch basis as follows: AAA = 1, AA+ = 2, AA = 3, AA− = 4, A+ = 5, A = 6, A− = 7, BBB+ = 8, BBB = 9, BBB− = 10, BB+ = 11, BB = 12, BB− = 13, B+ = 14, B = 15, B− = 16, CCC+ = 17, CCC = 18, CCC− = 19, CC = 20.

[19] We use OLS regression instead of logit or probit regression because it is problematic to estimate the coefficients of a nonlinear model with firm fixed effects, using logit or probit. Wooldridge (2002) notes that the maximum likelihood estimator in nonlinear panel data models with fixed effects is biased and inconsistent when T, the length of the panel, is small and fixed. Many prior studies (Naughton et al. 2018a, b) also use OLS when their dependent variables are binary variables.

[20] To save space, we present details on the construction of the two samples in Table OA1 of the online appendix.

[21] We also perform four additional analyses as discussed in the online appendix to further address endogeneity concerns: (1) using an instrumental variables approach (Table OA2), (2) controlling for the effects of factors that potentially drive options trading (Table OA3), (3) testing the impact threshold of a confounding variable (Table OA4), and (4) addressing the alternative explanation of managerial/firm behavior (Table OA5).

[22] For example, at the Chicago Board Options Exchange (CBOE), there are five criteria that must be met before the exchange will consider listing options for an underlying stock. The underlying stock must (1) be a registered National Market System stock; (2) have at least 7,000,000 publicly-held shares; (3) have at least 2,000 shareholders; (4) have a minimum trading volume of 2,400,000 shares in the past 12 months; and (5) have a sufficiently high price for a specific time. If a company meets these criteria, then it is eligible to have an option listing, but only the CBOE can decide which eligible companies will have listed options.

[23] To assess the robustness of our matching process, we also conducted propensity score matching, using alternative matching criteria, by selecting matching firms with the highest predicted probabilities of option listing and using the kernel matching method. The results using these alternative matching criteria are qualitatively similar, and, for brevity, we do not tabulate them.

[24] Our results are consistent if we construct our sample around the month of the options-listing event instead of the first month with non-zero options trading volume.

[25] We do not conduct a similar difference-in-difference analysis for rating-default predictability because the matched sample size is reduced to such a small number that there is little variation in the dependent variable $Default_{t+12}$. This sample has only 40 observations with default (0.70% of the full matched sample); the treatment subsample has 24 (0.84% of the treatment subsample) of these defaults and the control subsample has 16 (0.56% of the control subsample).

[26] In addition to these two approaches, we also construct two measures of private information in the options market and then test their respective relation with credit rating accuracy. Specifically, we follow Roll et al. (2010) and use the ratio of trading volume in out-of-the-money options to stock trading volume as our first measure of private information in option prices. The rationale is that out-of-the money options are more likely to be driven by private information because these options give informed traders higher leverage (Chakravarty et al. 2004, Pan and Poteshman 2006, Roll et al. 2010). Our second measure of private information in option prices is the proxy of Anderson et al. (2013) for informed options trading, that is, the t statistic for *Smirk* from an augmented Fama-French four-factor model that includes a one-day lag of the option smirk. We then replace *Vratio* with each of these two measures of private information in option prices and re-estimate our baseline models in Table 2. We find that credit rating accuracy improves with increases in private information in option prices, consistent with CRAs learning useful information from the option markets (untabulated for brevity).

[27] The coefficients on control variables are similar to those in Table 3, which, for brevity, we do not tabulate.

[28] In untabulated robustness checks, we repeat the analyses in Columns IV, V, and VI using an alternative specification of stock return volatility (i.e., standard deviations of stock returns over the fiscal year) and find similar results. To alleviate the influence of multicollinearity, we rerun the models in columns IV, V, and VI after including (one at a time) Ret_STD_t and its interaction, and YS_t and its interaction. We find positive and significant coefficients on the interactions of $Vratio_t$ with Ret_STD_t and YS_t, using the model in column V. Given that the within-firm variation for bond yield spread might be low due to the less active nature of the bond market, we also rerun the models in columns IV, V, and VI including (one at a time) Ret_STD_t and its interaction, and YS_t and its interaction, after replacing firm FE with industry FE. We find positive and significant coefficients on the interactions of $Vratio_t$ with Ret_STD_t

and $YS_{t.}$ in all model specifications except for the interaction of $Vratio_t$ with Ret_STD_t using the model in column IV.

[29] We also rerun all cross-sectional tests, including those in the text and online appendix, using the expanded model in column I of Table OA3 and find results similar to those reported in the paper (untabulated for brevity).

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